VALUE ADDED COURSE | PROJECT REPORT

AI IMAGE CLASSIFIER

Built with Google Teachable Machine

A no-code machine learning application for classifying fruit images

Submitted by	_____
Department	Artificial Intelligence and Data Science
College	_____
Academic year	_____

PROJECT DOCUMENTATION

01 | Executive Summary

This project demonstrates a practical use of artificial intelligence by building an image classifier with Google Teachable Machine. The application is designed to distinguish between three fruit categories—apple, banana and orange—using labelled example images. A user trains the model, supplies a new image through the webcam or image input, and reviews the predicted class and confidence score. The project illustrates a complete beginner-friendly machine-learning workflow without requiring custom code.

Project outcome: a trained prototype that can be tested with new fruit images. Prediction quality must be confirmed through actual testing; no accuracy figure is claimed in this report.

02 | Problem Statement

Sorting or identifying objects manually can be repetitive and may depend on a person's experience. This project explores whether a simple machine-learning model can classify common fruit images into predefined categories. The prototype focuses on three classes and provides a visual demonstration of how labelled data is used to train an AI model.

03 | Project Objectives

• Build an image classification prototype using a no-code machine-learning platform.

• Prepare labelled image examples for apple, banana and orange classes.

• Train the model and test it with images that were not used during training.

• Review predicted labels and confidence scores to understand model behaviour.

• Identify practical uses, limitations and possible improvements.

04 | Technology and Resources

Component	Purpose
Google Teachable Machine	Create, train and test the image model without writing code.
Browser and internet	Access the platform and complete model training.
Image dataset	Provide labelled examples for the three fruit categories.
Webcam or image upload	Supply images for testing the trained model.

05 | System Design and Workflow

• **Input:** labelled images representing apple, banana and orange.

• **Training:** Teachable Machine learns visual patterns from the labelled examples.

• **Model:** the trained image classifier maps an input image to the available classes.

• **Prediction:** a test image is processed and the model displays class probabilities.

• **Review:** the predicted class is compared with the known fruit to assess whether the result is correct.

Workflow
Collect images → Label classes → Train model → Test new images → Review predictions

06 | Implementation

1. Open the platform: Go to https://teachablemachine.withgoogle.com/ and select Get Started.

2. Create an image project: Choose Image Project and then Standard Image Model.

3. Set up classes: Create three classes named Apple, Banana and Orange.

4. Add training images: Upload or capture a varied set of images for each class. Around 30–50 examples per class is a useful starting point for a small demonstration, not a guarantee of accuracy.

5. Train the classifier: Select Train Model and wait for training to complete.

6. Test the model: Provide new images using the webcam or image input and note the predicted class.

7. Evaluate and document: Test multiple examples and record correct and incorrect predictions. Capture screenshots of the actual project and results.

07 | Testing and Evaluation

The classifier should be tested with images that were not used during training. Use different angles, backgrounds and lighting conditions. Record the true class, predicted class and whether the prediction was correct. If a confidence score is shown, record it as displayed by the platform. Do not use training performance alone as proof that the model works well on new images.

Test case	Actual class	Predicted class	Status
1	Apple	Fill after test	Correct / Incorrect
2	Banana	Fill after test	Correct / Incorrect
3	Orange	Fill after test	Correct / Incorrect

Optional metric: Test accuracy = (number of correct predictions ÷ total test images) × 100. Calculate this only after testing and counting the results.

08 | Potential Applications

• Fruit sorting and identification demonstrations.

• Plant or leaf category recognition after training on suitable images.

• Waste classification prototypes for educational use.

• Visual quality-control demonstrations in manufacturing.

• Student projects that introduce AI and machine learning concepts.

09 | Benefits and Limitations

Benefits

• No custom programming is needed to create the initial model.

• The interface makes the training and prediction process easy to demonstrate.

• The model can be improved by adding varied and correctly labelled examples.

Limitations

• Predictions can be affected by image quality, lighting, background and class imbalance.

• The model is intended to recognize only the classes provided during training.

• It may produce incorrect predictions; confidence scores do not guarantee correctness.

• A small classroom prototype is not a substitute for a professionally validated commercial system.

10 | Future Enhancements

• Add more fruit categories and increase the diversity of the dataset.

• Test the model using a larger, separate test set.

• Export the model and integrate it into a simple web or mobile interface, if required.

• Compare performance under different lighting and background conditions.

11 | Project Evidence

Insert genuine screenshots from the completed Teachable Machine project in the spaces below or append them as separate pages: (1) class setup and uploaded examples, (2) training screen, and (3) prediction for a new test image. Captions should describe what each screenshot demonstrates.

Screenshot 1 — Dataset classes

Paste screenshot here

Screenshot 2 — Model training

Paste screenshot here

Screenshot 3 — Test prediction

Paste screenshot here

12 | Conclusion

The project presents a working approach for creating an image classifier with Google Teachable Machine. By organizing fruit images into labelled classes, training the model and testing it with new examples, the project demonstrates the main stages of a machine-learning application. Final performance should be reported using the actual test results collected during implementation.

13 | Official Resource

Google Teachable Machine: teachablemachine.withgoogle.com